Problem Classes: LP, QP, SOCP, SDP

Convex Optimization, Take-home Quiz 4

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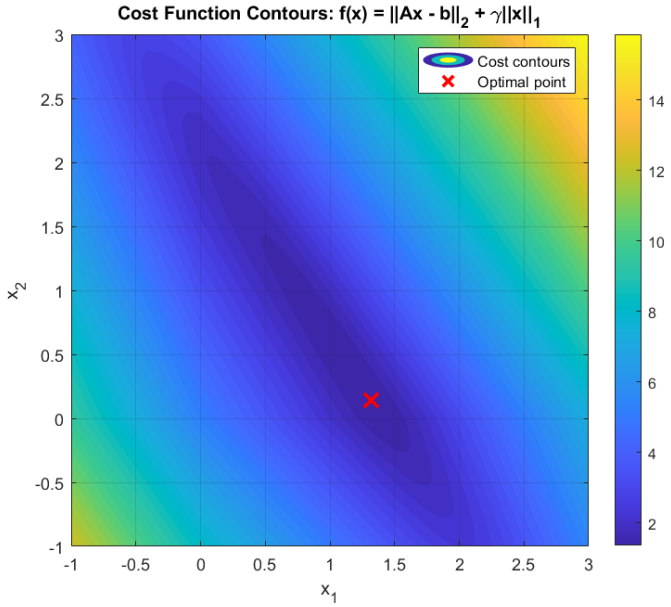
The code for question 7 is `Quiz_Code.m` (YALMIP).

The quiz asked me to decide whether problems are convex and to rewrite them in a standard form:

1. **Optimality in a QP.** At the optimum $x^\star$ of a convex QP over a feasible set P, $-\nabla f(x^\star)$ is a normal to P at $x^\star$: the level set of f just touches the boundary of P.
2. **Two problems that look hard.** Minimising $\|x\|^2$ outside two discs is not convex, because the feasible set is the complement of a union of discs. Minimising the Gaussian CDF of $c^T x$ subject to $Ax \le b$, $Hx = g$ is quasi-convex but not convex.
3. **A sum of $(|a_i^T x| - 1)^2$.** This objective is not convex. Introducing $p_i = |a_i^T x|$ and relaxing gives a QP, a QCQP or an SOCP.
4. **A piecewise-affine objective.** I split $c^T x + \max(a_2^T x + b_2,\ a_3^T x + b_3)$ into two cases, each an LP.
5. **Travel time across two media.** The travel time $T(x)$ from one medium to another (a Snell's-law setting) is convex, which I showed with its second derivative. I extended this to N layers by checking the Hessian.
6. **The Bellman equation as an LP.** The value function solves $\min_v \sum_s \mu(s)v(s)$ subject to $v(s) \ge \sum_{s'} P(s'|s,a)[r + \gamma v(s')]$ for all s, a.
7. $\min_x \|Ax-b\|_2+\gamma\|x\|_1$ **as an SOCP,** with $t \ge \|Ax-b\|_2$ and $-u \le x \le u$. Solved numerically below.
8. $\min \|Ax-b\|_p$**.** $p = 2$ with complex data becomes a real least-squares problem in $z = [\Re x; \Im x]$. $p = \infty$ and $p = 1$ become SOCPs.
9. **Bounding a polynomial.** Finding $\inf_t P(t)$ of a polynomial with interval bounds becomes an SDP through a Gram-matrix (sum-of-squares) certificate $P(t) - \gamma = v(t)^T Q v(t)$, $Q \succeq 0$.

Question 7, solved numerically

`Quiz_Code.m` solves question 7 for $A = \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 2 \end{bmatrix}$, $b = (2, 3, 4)$ and $\gamma = 0.5$, then draws the cost contours with the optimum marked.


Cost Function Contours: f(x) = ||Ax - b||2 + γ||x||1
Cost contours
Optimal point
x1
x2
3
2.5
2
1.5
1
0.5
0
-0.5
-1
-1
-0.5
0
0.5
1
1.5
2
2.5
3
14
12
10
8
6
4
2